



National University of Sciences and Technology (NUST)
School of Electrical Engineering and Computer Science

Department of Computing

CS220: Database Systems

Class: BSAI – 2A

Lab 06: DDL and Constraints

CLO-2: Formulate SQL queries to retrieve information from a relational database.

CLO 3: Build SQL (DDL &DML) queries to retrieve information from a relational database.

Date: 05 March 2026

Time: 09:45 - 11:55



Introduction

This lab will focus on the DDL (Data Definition Language) part of SQL. It discusses how to create relations and discusses different types of constraints defined in SQL. Constraints are a specific set of rules defined for SQL objects. In this lab, you will implement constraints and then observe how a constraint ensures that valid data is inserted into the relations.

Objectives

After completing this lab, you should be able to do the following:

- Create relations in a database
- Create constraints, including NOT NULL, Unique Key Constraint, Primary Key, and Foreign Key

Tools/Software Requirement

- MySQL Community Server 5.6
- MySQL Workbench 6.1

Description

Creating Tables

1) The CREATE TABLE Statement

First of all, create a new schema/database or use an existing one to create tables in it. Create tables/relations to store data by executing the SQL CREATE TABLE statement. This statement is one of the data definition language (DDL) statements. DDL statements are a subset of SQL statements used to create, modify, or remove database structures. These statements have an immediate effect on the database, and they also record information in the data dictionary.

To create a table, a user must have the CREATE TABLE privilege and a storage area in which to create objects. The syntax to create a table is as follows:

```
CREATE TABLE [schema.] table  
  (column datatype [DEFAULT expr]  
    [column_constraint],  
    ...
```



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[table_constraint][,...]);

In the syntax:

schema	is the database in which to create table
table	is the name of the table
DEFAULT expr omitted in the INSERT	specifies a default value if a value is statement
Column	is the name of the column
datatype	is the column's data type and length
column constraint	is the constraint to apply on table

Example:

Create a table employees and departments.

```
CREATE TABLE employees(  
    employee_id INT,  
    first_name VARCHAR(20),  
    job_id INT NOT NULL,  
    dept_id INT,  
    CONSTRAINT emp_emp_id_pk  
        PRIMARY KEY (EMPLOYEE_ID),  
    CONSTRAINT emp_dept_id_fk  
        FOREIGN KEY (dept_id) references departments (department_id)  
);
```

```
CREATE TABLE departments(  
    department_id INT,
```



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```
department_name VARCHAR(30),  
  
location VARCHAR(30),  
  
job_id INT,  
  
CONSTRAINT dept_dept_id_pk  
  
PRIMARY KEY (department_ID));
```

Notice the sequence of creating the tables. Due to the referential integrity constraint, the sequence of execution of the table creation queries of employees and departments is important.

2) AUTO-INCREMENT values

In case you want a column to auto_increment (mostly ID columns), you can use the following syntax:

```
CREATE TABLE departments(  
  
    department_id INT AUTO_INCREMENT,  
  
    department_name VARCHAR(30),  
  
    location VARCHAR(30),  
  
    job_id INT,  
  
    CONSTRAINT dept_dept_id_pk  
  
    PRIMARY KEY (department_ID));
```

3) DEFAULT value

For a value of a column that is not explicitly supplied while inserting values into a table, a default value can be used. The default value for a column is specified at the time of creation of table.

```
CREATE TABLE departments(  
  

```



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```
department_id INT,  
  
department_name VARCHAR(30),  
  
location VARCHAR(30),  
  
job_id INT default 10,  
  
CONSTRAINT dept_dept_id_pk  
  
PRIMARY KEY (department_ID));  
  
);
```

Lab Task

Schema to be implemented:

Student (snum: integer, sname: char(30), major: char(25), level: char(2))

Faculty (fid: integer, fname: char(30), deptid: integer)

Class (cname: char(40), meets_at: char(20), room: char(10), fid: integer | fid REFS Faculty.fid)

Enrolled (snum: integer, cname: char(40) | snum REFS student.snum, cname REFS class.name)

Table Name	Primary Key	Foreign Key	Reference Table
Student	Snum		
Faculty	Fid		
Class	Cname	fid	faculty
Enrolled		snum, cname	Student, Class



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The University schema is given in the figure above. Using this schema, perform the following task:

Task: Create a University database and the corresponding tables/relations in the University schema using the DDL commands. Ensure only the following constraints are specified during the table creation time:

- Primary Key
- Foreign Key
- Not Null
- Unique

PK values are given in the figure. sname attribute should not be null, while deptid should always be unique. Assign valid names to all the constraints.

Deliverables

Save all queries and their results in the word document that you are executing. Upload this document to LMS.

Student Table:

Use university;

```
Create table Student3(  
snum INT primary key,  
sname char(30) not null,  
major char(25),  
level1 char(2)  
);
```

Result Grid:

Result Grid				
	snum	sname	major	level1
*	NULL	NULL	NULL	NULL



Class Table:

```
Create table Class3(  
  cname char(40) primary key,  
  meets_at char(20),  
  room char(10),  
  fid INT,  
  Constraint fk_Class_Faculty foreign key(fid) references Faculty3(fid)  
);
```

Result Grid:

Result Grid			
	fid	fname	deptid
	NULL	NULL	NULL

Faculty Table:

```
Create table Faculty3(  
  fid INT primary key,  
  fname char(30),  
  deptid INT  
);
```



Result Grid:

Result Grid				
	cname	meets_at	room	fid
*	NULL	NULL	NULL	NULL

Enrolled Table:

Create table Enrolled3(
snum INT,
cname char(40),

Constraint fk_Enrolled_Student foreign key(snum) references Student3(snum),
Constraint fk_Enrolled_Class foreign key(cname) references Class3(cname)
);

Result Grid:

Result Grid		
	snum	cname



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Full Result Table display:

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	snum	sname	major	level1	cname	meets_at	room	fid	snum	cname	fid	fname	deptid
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Result 6

×

Read Only

Database created:

